Survival Analysis Overview

Survival Analysis

Study time until an **event** occurs (e.g. death, disease progression, hospital discharge).

Methods

- Kaplan-Meier Curve (Descriptive)
- Log-Rank Test (Hypothesis Test)
- Cox Proportion Hazards Regression (Regression)

Data Format

- **Time:** Time until the event occurred or the observation was censored
- **Status:**
  - 0 = Censored (event not observed during follow-up)
  - 1 = Event occurred

Survival Analysis accounts for censored observations.

Demo

Health and Retirement Study (hrs) dataset

Knowledge Check

True or False:

We must remove all censored observations before beginning survival analysis.

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